Designing a Bunny Rally Virtual Reality Environment to Support Panksepp's PLAY System

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Abstract

Virtual reality (VR) can support emotion elicitation by combining immersion, interaction, and agency. Unlike conventional methods involving static images or videos, VR enables participants to engage actively with emotionally meaningful events, potentially strengthening emotional engagement. PLAY, one of Jaak Panksepp's primary-process emotional systems, is associated with social joy, positive affect, and playful interaction.

We designed a Virtual Environment (VE) to support PLAY by adapting an affective-VR design methodology. We extended this methodology with a social-reciprocity design component in which the participant and a non-player character (NPC) engage in contingent, back-and-forth interaction. In this interaction, participant actions influence the NPC's subsequent behaviour, while the NPC's responses reshape the ongoing interaction. The resulting design space included semantic and affective framing, sensory properties, dynamism, interaction and agency, and NPC behaviour.

To evaluate the design, we conducted a within-participant study (n = 26) in which participants completed a tutorial VE which served as the baseline followed by an affective VE designed to elicit PLAY. Self-report measures of affect, PLAY-related emotional states, and presence were collected before and after the tutorial VE and after the affective VE. The results indicated that Post-affective scores exceeded the selected reference points for valence, presence, and co-presence. However, after correcting for multiple comparisons, no significant increases were observed in the primary affective or PLAY-related measures from the post-tutorial to the post-affective assessment. The tutorial also produced changes in participants' emotional states. Therefore, changes observed after the affective VE cannot be attributed solely to its affective content.

Keywords: virtual reality, emotion elicitation, PLAY system, Panksepian affective neuroscience, affective virtual environments, social reciprocity

1 Introduction

Emotions influence how people perceive events, make decisions, regulate their behaviour, and interact with others. Controlled emotion elicitation is therefore important in psychology, neuroscience, and human-computer interaction because it allows researchers to examine the effects of emotional states [19].

Traditional elicitation methods often use passive stimuli, such as images, videos, or sounds. Virtual reality (VR) provides a more active alternative by combining multisensory presentation with interaction, presence, and agency. Participants can move, manipulate objects, and respond to events within the environment rather than simply observe them. However, VR does not automatically elicit a specific emotion, the response depends on the emotional meaning, sensory design, interaction, and context of the experience [3, 8, 15, 21].

This study focuses on PLAY, one of Jaak Panksepp's primary-process emotional systems. PLAY is associated with social joy, positive affect, laughter, and active playfulness. Although it is sometimes described as "social joy," PLAY is not synonymous with general happiness. It refers specifically to active, socially directed interaction. Panksepp's three-tiered model proposes that primary processes generate basic emotional urges, secondary processes involve learned associations, and tertiary processes involve higher-order reflection. These levels interact bidirectionally, meaning that activation at one level can influence the others [6, 13, 20].

The affective VE draws on familiar conventions of bat-and-ball games, including turn-taking, throwing, hitting, and catching. It uses these learned interaction patterns to support primary-process PLAY. Since PLAY is socially directed, the VE also incorporates reciprocal interaction through a non-player character (NPC). The bunny initiates actions, responds to the participant's performance, and continues the interaction according to the outcome. This contingent back-and-forth structure distinguishes the VE from one based only on positive visual or auditory stimuli.

Designing this experience requires a systematic method for translating affective theory into virtual-environment features. We therefore adapt an affective-VR design methodology shown in Figure 1 to account for the social reciprocity and contingent interaction associated with PLAY. The resulting design combines semantic and sensory elements, dynamism, agency, and NPC behaviour.

The main research question is: How can an affective-VR design methodology be adapted to create a Virtual Environment that supports the elicitation of Panksepp's PLAY system? To address this question, we designed a bunny-rally environment and evaluated it in a within-participant study. Participants completed a tutorial environment followed by

the affective VE, and reported their affective states, PLAY-related responses, presence, social presence, and perceived co-presence.

This paper makes three contributions. First, it presents a PLAY-specific adaptation of an affective-VR design methodology. Second, it describes a design space combining semantic, sensory, dynamic, interactive, and reciprocal elements. Third, it evaluates the resulting VE and examines whether the observed responses are consistent with the elicitation of PLAY.

2 Related Work

2.1 VR for Emotion Elicitation

Emotion elicitation methods include visual stimuli, music, autobiographical recall, imagery, and situational procedures. Although these methods can produce emotional responses, many involve passively observing or imagining an emotional situation [19]. VR offers a more active and embodied alternative by allowing participants to move, manipulate objects, make decisions, and receive responses from the environment. These features can increase the contextual richness and personal relevance of the eliciting experience [8, 15, 21].

VR nevertheless does not guarantee the elicitation of a particular emotion. Responses depend on how the environment's sensory cues, interaction, narrative context, and feedback are combined. For example, interactivity has been shown to influence joy in VR, although its effects may depend on participants' prior mood [3]. Existing research has focused mainly on general positive affect, joy, anxiety, fear, and relaxation. Less attention has been given to designing for Panksepian primary-process systems, particularly PLAY, or to incorporating reciprocal interaction with an NPC. This study addresses this gap by adapting an affective-VR methodology for socially contingent PLAY.

2.2 Presence, immersion, agency, and social interaction

Immersion describes the technological capacity of a system to provide continuous multisensory input, whereas presence describes the subjective feeling of being located within the Virtual Environment [15, 24]. Greater immersion generally increases presence, but presence is also influenced by interactivity, responsiveness, and emotional meaning. In emotionally engaging environments, these experiential qualities may contribute to presence beyond graphical realism alone [8].

Agency refers to the perception that one's actions meaningfully influence the environment and its events. In an affective VE, agency can be supported through meaningful participant actions and contingent NPC responses. Previous research has linked interactivity and perceived agency with engagement, joy, telepresence, co-presence, and social presence [3, 12].

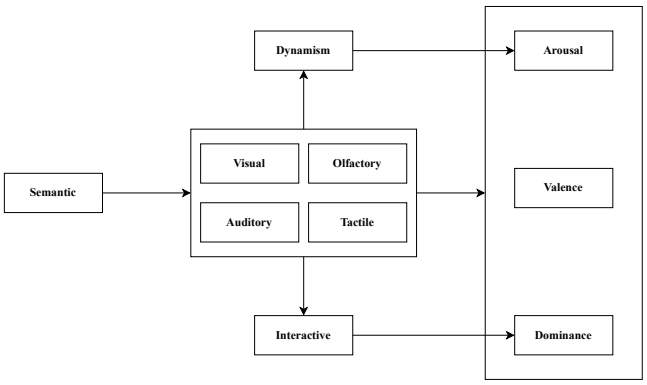


Figure 1. Dozio et al.'s affective VE design procedure

Co-presence refers to the feeling that another entity is present in the same virtual space, whereas social presence concerns the extent to which that entity is perceived as a socially meaningful interaction partner [12]. These measures indicate perceived social connection, but they do not by themselves demonstrate reciprocity or PLAY. Reciprocity requires contingent back-and-forth responses in which the participant's actions influence the NPC's subsequent behaviour. This distinction motivates the present study's extension of affective-VR design to include social reciprocity.

2.3 Affective VR design methodologies

Affective-VR design methodologies provide a structured way to select and combine environmental features that support a target emotional experience [9]. These features include semantic elements, sensory cues, dynamism, and interaction. Semantic elements communicate the meaning of the environment, sensory cues influence the user's affective response, dynamism concerns movement and change, and interaction concerns the relationship between the participant's actions and the environment's responses.

The methodology by Dozio et al in Figure 1 commonly describes affect using valence, arousal, and dominance. Valence represents pleasantness, that is how good or bad one feels. Arousal represents how calm or excited one feels, and dominance represents perceived control or influence [9]. In this study, PLAY is operationalised as involving high valence, high arousal, high dominance, and reciprocal interaction between the participant and the NPC [13, 20].

Although Dozio et al.'s methodology provides a useful basis for designing affective Virtual Environments, it was developed for discrete emotions such as happiness and does not explicitly address Panksepp's PLAY system or reciprocal NPC interaction. The present study adapts the methodology to incorporate these socially contingent characteristics.

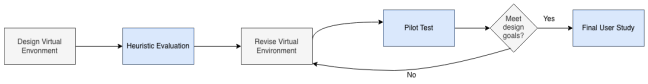


Figure 2. Iterative virtual-environment design and evaluation process

3 Design

3.1 Overall design methodology

As discussed in Section 2.3, PLAY behaviours can be operationalised as involving high valence, high arousal, and high dominance. In addition to these dimensions, reciprocal social interaction was treated as a central design requirement for supporting PLAY. Reciprocity occurs when the participant's actions influence the NPC's subsequent behaviour, which then creates an opportunity for the participant to respond again.

The VE was designed as a bunny-rally interaction in a forest setting on a bright, sunny day. It involved an anthropomorphic bunny that throws acorns to the participant, who is invited to hit them back for the bunny to catch using a bat. This interaction was chosen because it includes turn-taking, energetic movement, user control, and back-and-forth interaction with the NPC. These features were intended to support the emotional qualities associated with PLAY.

The VE was designed as a bounded but explorable environment to provide the participant with a sense of agency while preventing unintended behaviour. For example, restrictions were used to prevent the participant from wandering into the forest, becoming lost, or dropping the bat and being unable to find it. These constraints were intended to support meaningful interaction while maintaining a clear focus on the bunny-rally activity.

The process of designing the VE involved several iterations, as shown in Figure 2. A heuristic evaluation was conducted before the user study by two evaluators with relevant experience, the supervisor and a Master's student. Nielsen's heuristics were selected to identify general usability problems, such as unclear feedback, inconsistent design, and a lack of user control. Sutcliffe and Gault's adaptation was also used because it addresses challenges specific to virtual-reality applications, including navigation, interaction, viewpoint, and presence [22]. Using both sets of heuristics allowed us to identify problems that could interfere with participants' interaction, agency, presence, or emotional response. One issue was an inconsistency in environmental motion cues, the grass swayed while the trees remained static. The evaluation therefore informed revisions to the VE before user testing.

3.2 Design elements

Table 1 summarizes the design elements used in the VE and their intended contributions. These elements were not treated as independent features with fixed emotional effects. Instead, they were selected to work together to create a coherent emotional and social experience. A visualization of some of these elements is shown in Figure 4.

The design was intended to produce positive arousal rather than simply a calm or pleasant emotional state. The playful activity, movement, and changing events created opportunities for excitement and anticipation, while the positive semantic and sensory cues were intended to maintain the pleasant quality of the experience. This distinction is important because high arousal can also occur during negative emotions such as fear or anger. Therefore, arousal was supported through energetic activity, while valence was supported through the positive setting, sounds, colours, and character design.

The VE also involved a trade-off between agency and guidance. The path, footsteps, glowing bat, and hand symbol helped participants understand what actions were possible, as shown in Figure 5. These cues were intended to reduce confusion without removing meaningful control from the participant.

The most important design consideration was the relationship between interaction and reciprocity. Guidance cues helped participants understand how to begin the experience, but reciprocity required the participant's actions to influence the NPC's subsequent responses, creating further opportunities for the participant to act. To support this interaction, an approachable anthropomorphic bunny was used, as shown in Figure 3. Its soft cream colour and long ears were intended to make it appear inviting. The use of human-like behaviour was informed by findings that perceived agency and the representation of virtual interactants can influence co-presence and social presence [12]. However, a non-human animal character was chosen to reduce the risk of the discomfort associated with near-human realism in the uncanny valley[7]. The bunny's appearance supported approachability, while its contingent responses implemented reciprocity. This interaction distinguished the VE from one relying only on positive visual and auditory stimuli and provided the primary social mechanism for supporting PLAY.

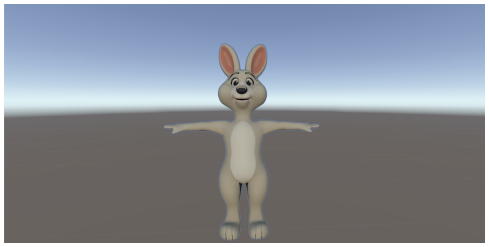


Figure 3. Bunny

The International Affective Digitized Sounds (IADS) ratings were used as a guide for selecting auditory stimuli. More specifically, birdsongs were chosen as they are associated

Table 1. Design elements supporting PLAY

Element	Implementation and intended role
Semantic	Sunny forest, bunny, fruits, and flowers; establishes positive valence and a socially inviting play context.
Sensory	Bright colours, birdsong, bunny vocalisations, and interaction sounds; reinforces positive valence and supports heightened arousal.
Dynamism	Moving acorns, throwing, catching, and bunny movement; sustains arousal, anticipation, and active engagement.
Agency	Participant controls movement, hand choice, aiming, and hitting; supports perceived dominance through meaningful control over play.
Reciprocity	Bunny throws acorns, responds to outcomes, and continues the interaction; supports contingent turn-taking and socially directed PLAY.

with positive valence in the database[25]. However, participants may respond to these sounds differently depending on their prior experiences, the surrounding visual context, and the NPC interaction. Similarly, the design elements were expected to support PLAY but could also produce general happiness, engagement, curiosity, or excitement. The user study was therefore needed to determine whether the combined design produced the intended affective and social responses.

4 Virtual Environment Implementation

4.1 Unity environment and technical setup

The VE was developed using Unity 6.2 and the XR Interaction Toolkit. Participants experienced the VE using an HTC Vive 2018 headset and handheld controllers. Teleportation was implemented to allow participants to move through the forest while reducing the need for physical walking. Snap turning in 25-degree increments was used instead of smooth continuous turning to reduce the likelihood of motion sickness. The bunny animations were sourced from Mixamo, while environmental assets, including the trees, barrel, grass, and bat, were obtained from the Unity Asset Store. Audio clips were sourced from Pixabay, and additional sounds were recorded for the VE.

4.2 Virtual Environment and object implementation

As mentioned in Section 3, the Virtual Environment is set in a forest. The user begins approximately 100 metres from the interaction location, with a gravel path guiding them towards the interaction area. Once the user arrives, they encounter a barrel with a bat placed on top, signaling that the bat should be picked up to begin the interaction. Figure 6 shows the floor plan for this Virtual Environment.

The interaction experience involves two main objects: an acorn and a bat. Hits and catches were detected using invisible boundaries around the virtual objects. When the bat came into contact with an acorn during a swing, the system registered a hit. The direction of the acorn after contact was then checked to determine whether it was travelling towards the bunny and in that case would be considered a successful hit. An invisible catch area around the bunny detected whether the acorn entered a suitable position for catching. If the acorn entered this area, the catch was recorded and the bunny played its catching animation and celebratory sound. If the acorn travelled in the wrong direction or did not enter the catch area, the attempt was recorded as unsuccessful and the bunny retrieved the acorn, this acorn is referred to as a resolved acorn.

4.3 Interaction Progression

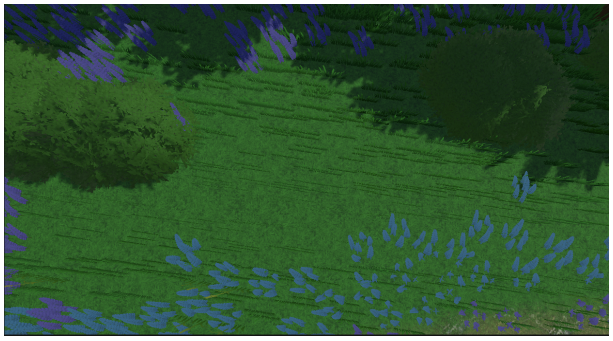
At the beginning of the experience, the participant faced a path while the bunny ran ahead towards the interaction area. The participant followed the path and picked up the glowing bat placed on a barrel, which initiated the bunny-rally interaction. The bunny then threw an acorn towards the participant. A successful hit caused the bunny to catch the acorn, jump, and play a “Yay” sound. If the acorn was missed or hit away from the bunny, the bunny retrieved it before continuing the interaction. The interaction lasted 7.5 minutes, and its progression is shown in Figure 7.

Each stage had different base settings, including acorn speed, throw interval, bunny movement, spread angle, and catch requirements, as summarised in Table 2. A stage was completed once the required number of catches had been achieved, after which the Stage Manager loaded the settings for the next stage.

Adaptive difficulty was included to maintain a balance between challenge and successful participation. If the interaction was too difficult, repeated failures could lead to frustration and disengagement, reducing the participant’s opportunity to experience reciprocal interaction. If it was too easy, the activity could become predictable and less stimulating, potentially reducing arousal and engagement. The Stage Manager therefore evaluated performance after every four resolved acorns. Accuracy of 45% or lower reduced the difficulty by slowing the acorns, increasing the interval between throws, and reducing bunny movement. Accuracy of 75% or higher, together with a catch streak of at least three, increased the difficulty by speeding up the acorns, decreasing the interval between throws, and increasing bunny movement. If neither condition was met, the difficulty remained unchanged. These adjustments were intended to maintain engagement while preserving opportunities for successful back-and-forth interaction. They occurred within the current stage and did not reset the participant’s catch progress. These thresholds were practical design rules for this prototype rather than validated boundaries. Because adaptive



(a) Bright blue sky

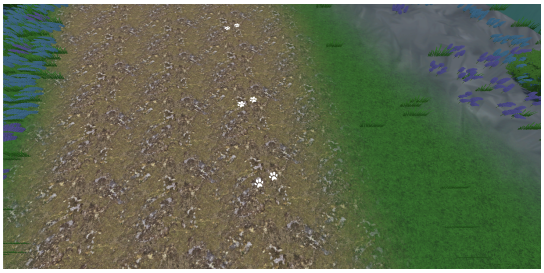


(b) Highly saturated green grass

Figure 4. Environment colours



(a) Glowing bat



(b) Footsteps

Figure 5. Environment affordances

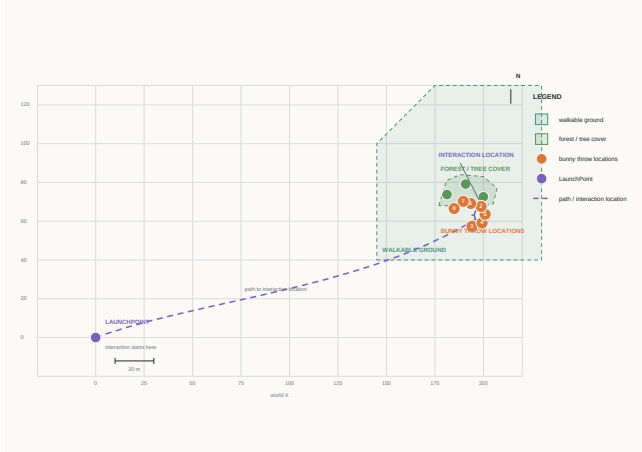


Figure 6. Floor plan

difficulty meant that participants could experience slightly different interaction conditions, the adjustments should also be considered when interpreting the emotional results

4.4 NPC behaviour and feedback

At its base setting,the bunny throws acorns every 2.75 seconds. This interval could be increased or decreased by the adaptive difficulty system.The bunny also produces several vocalisations, including "Yay," "Come on, you've got this," and "Yoohoo, over here." It also has several animations, including throwing, dancing, running, catching, waving, retrieving an acorn from the ground, and an idle animation between throws. The bunny's behaviour was dependent on the participant's actions in order to create reciprocity and a sense of agency.

5 Experiment design

A within-subjects study (n=26) was conducted to evaluate the effectiveness of the affective Virtual Environment in eliciting the PLAY system. The independent variable was the Virtual Environment condition (Tutorial vs. Affective Environment). The primary dependent variables were self-reported measures of emotional affect and spatial/social presence, quantified using a suite of standardized questionnaires.

5.1 Apparatus

The study was conducted in person in Experiment Room 1 in the Computer Science building. A desktop computer running on Windows 10 was used to run the experiments. The head-mounted display used was the HTC Vive 2018 headset and a set of compatible HTC controllers. Responses from the questionnaires were collected on paper as we established it as the most appropriate medium allow participants to see the full suite of questionnaires avoiding a sense of overwhelm.

Table 2. Base settings for each interaction stage

Stage	Catches required	Acorn speed (m/s)	Throw interval (s)	Spread angle (°)	Bunny movement	Movement frequency
1	4	15	2.75	5	No	Every 5 throws
2	4	15	2.75	5	Yes	Every 2 throws
3	7	15	2.75	5	Yes	Every throw
4	4	15 (first acorn), 11 (second acorn)	2.75	5	Yes	Every throw

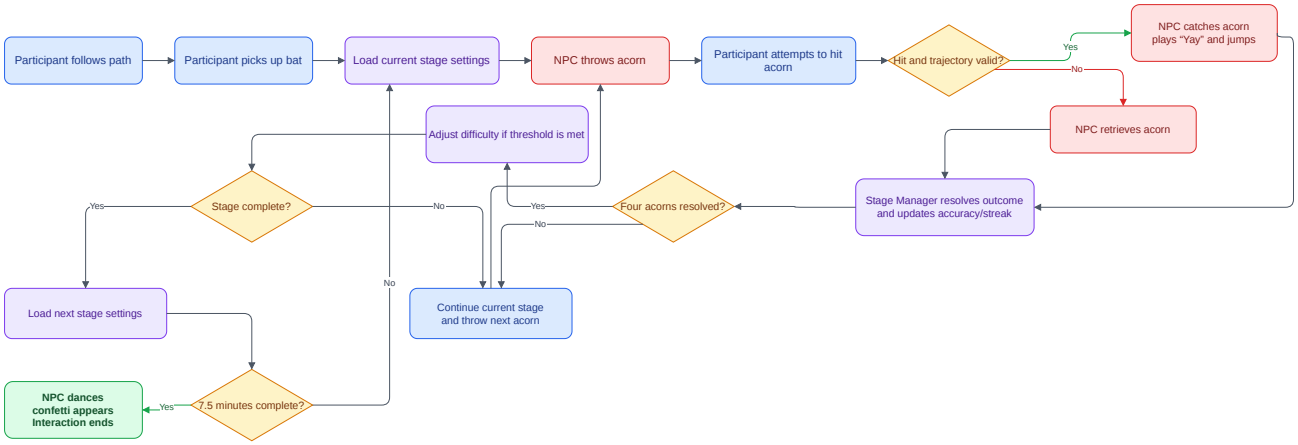


Figure 7. Interaction flow between the participant, NPC, and Stage Manager

Each participant who took part in the study was automatically entered into a draw for 1 of 3 R300 Takealot vouchers as a form of compensation for their participation. This study was approved by the UCT Inter-faculty Research Ethics Committee reference number IFHREC/02979/2026.

5.2 Participants

A total of 26 participants aged 18 years or older were recruited through university and community networks using snowball sampling. Snowball sampling was used because it provided a practical and efficient way to recruit eligible adult participants through existing university and community networks within the project's time and resource constraints. Participants were a mix of undergraduate students from different faculties and some from outside the university. Interested individuals completed an online screening form that collected information about prior VR experience, history of motion sickness, diagnosed psychiatric conditions, chronic stimulant-medication use, and depressive or anxiety disorders. Prior VR experience was recorded as a participant characteristic, while the remaining factors formed the exclusion criteria. These criteria were selected because psychiatric conditions and stimulant medication may influence emotional reactivity, while depression and reduced social playfulness could affect participants' responses to the PLAY-inducing interaction [11, 13, 18]. A history of motion sickness was also an exclusion criterion because it may increase susceptibility to cybersickness during VR exposure [14]. Participants additionally completed the Beck Depression Inventory-II (BDI-II), those scoring above the study's prespecified cut-off of 20 were excluded because elevated depressive symptoms may reduce social-play behaviour [1, 11].

5.3 Measures

Each questionnaire was selected to measure a distinct component of the participant's experience. Although physiological measures can provide continuous indicators of arousal, they were not used because the accuracy of devices depends on correct sensor placement, and placement errors could introduce measurement noise [17]. Self-report measures were therefore used because they directly capture participants' perceived emotional and experiential states.

The Self-Assessment Manikin (SAM) was selected because it measures valence, arousal, and dominance using brief pictorial nine-point scales[4]. These dimensions correspond directly to the affective-VR framework used in this study and allow the results to be compared with the normative ratings of the International Affective Picture System (IAPS) and IADS databases. The PANAS-X Joviality subscale was included to assess positive affect associated with cheerfulness, enthusiasm, and social enjoyment [23]. It was selected instead of a general positive-affect measure because these

qualities are more closely related to the positive emotional characteristics of PLAY. However, joviality does not specifically demonstrate that PLAY was elicited.

The Affective Neuroscience State Scales (ANSS) are a self-report measure of Pankseppian emotional systems. In this study, ANSS scores were used to assess PLAY and related emotional states. Because the instrument is still undergoing validation and uses a recall period, results were treated as exploratory. Finally, the iGroup Presence Questionnaire(IPQ) was used to assess general presence, while the co-presence and social-presence measures assessed whether the participant experienced the NPC as a socially meaningful interaction partner[2, 12]. The IPQ uses a −3 to +3 scale, making zero the neutral midpoint. We tested whether participants' presence scores differed from zero. This midpoint is a neutral reference, not a validated cutoff for high presence[2].

5.4 Procedure

The following procedure was followed during each session. Participants first completed a tutorial environment consisting of a blue room with a red box on the floor marking the permitted interaction area. The tutorial introduced the basic mechanics required for the affective Virtual Environment, including teleportation, snap turning, picking up objects, and throwing objects. The tutorial environment was provided by James Burness, a Master's student.

Introduction (∼3 minutes) – Participants completed the consent form and received an overview of the study procedure.

Pre-tutorial questionnaire suite (∼7 minutes) – Participants completed the SAM, PANAS-X, and ANSS questionnaires before entering the tutorial environment.

Tutorial Virtual Environment (∼5 minutes) – Participants completed the tutorial VE and practised teleportation, turning, object manipulation, and throwing.

Post-tutorial questionnaire suite (∼9 minutes) – After completing the tutorial VE, participants completed the SAM, PANAS-X, ANSS, and IPQ questionnaires.

Affective Virtual Environment (∼7.5 minutes) – Participants then experienced the affective VE for approximately 7.5 minutes.

Post-affective questionnaire suite (∼10 minutes) – Immediately after the affective VE, participants completed the SAM, PANAS-X, ANSS, IPQ, co-presence, and social-presence questionnaires.

Debriefing (∼3 minutes) – After completing the final questionnaires, the researcher debriefed participants about the study's purpose and research question. At this stage, participants were informed that the study focused on the PLAY emotional system. They received either a printed copy of the debriefing form or a digital copy accessed through a QR code.

5.5 Hypotheses

We tested whether the affective VE produced:

(H1) high valence, arousal, dominance, presence, co-presence, and social presence

(H2) increases in SAM, PANAS-X, IPQ and ANSS scores relative to the tutorial

(H3) higher PANAS-X Joviality than the other PANAS-X subscales and higher ANSS PLAY than the other ANSS subscales

We also tested whether the tutorial itself changed emotional-state measures.

6 Results

All 26 participants completed the study. Complete-case analyses used $n = 26$, except ANSS PLAY ($n = 25$), PANAS-X ($n = 24$), IPQ ($n = 25$), and baseline ANSS CARE ($n = 25$), no imputation was used. Normality was assessed for each paired change score using the Shapiro–Wilk test and Q–Q plots. Paired-samples t-tests were used when the change scores were approximately normally distributed, and Wilcoxon signed-rank tests were used when meaningful departures from normality were observed. All tests were two-sided, and Holm adjustments were applied within the relevant comparison family. The significance level was $\alpha = .05$.

6.1 Primary questionnaire comparison

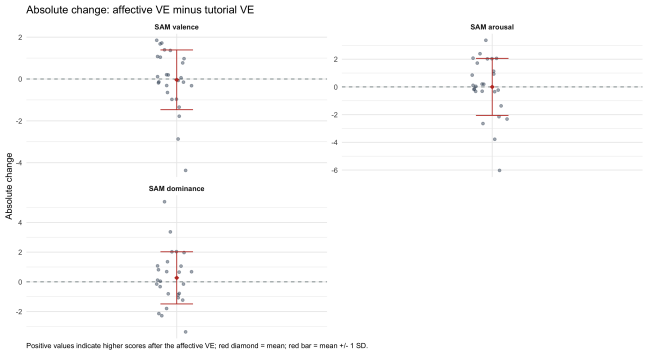
The primary test of H2 compared post-affective with post-tutorial scores. Positive values indicate higher scores after the affective VE. SAM valence changed from $M = 7.27$ ($SD = 1.54$) to $M = 7.23$ ($SD = 1.68$), $\Delta M = -0.04$, $W = 70.5$, $p_{\text{Holm}} = 1.000$. SAM arousal was unchanged, $\Delta M = 0.00$, $W = 72.0$, $p_{\text{Holm}} = 1.000$, while SAM dominance increased descriptively by 0.27 points, $t = 0.78$, $p_{\text{Holm}} = 1.000$.

No PANAS-X subscale showed a significant post-affective increase after Holm correction. PANAS-X Joviality changed from $M = 3.64$ ($SD = 0.66$) to $M = 3.64$ ($SD = 0.78$), $\Delta M = -0.01$, $t = -0.04$, $p_{\text{Holm}} = 1.000$. ANSS PLAY changed from $M = 3.58$ ($SD = 0.83$) to $M = 3.61$ ($SD = 0.89$), $\Delta M = 0.03$, $t = 0.28$, $p_{\text{Holm}} = 1.000$, and no other ANSS subscale showed a significant change. IPQ presence increased from $M = 0.35$ ($SD = 0.77$) to $M = 0.74$ ($SD = 0.80$), $\Delta M = 0.40$, $t = 2.39$, unadjusted $p = .025$, $p_{\text{Holm}} = .597$. The complete measure-level results, including all PANAS-X and ANSS subscales, are given in Appendix Tables 3 - 6.

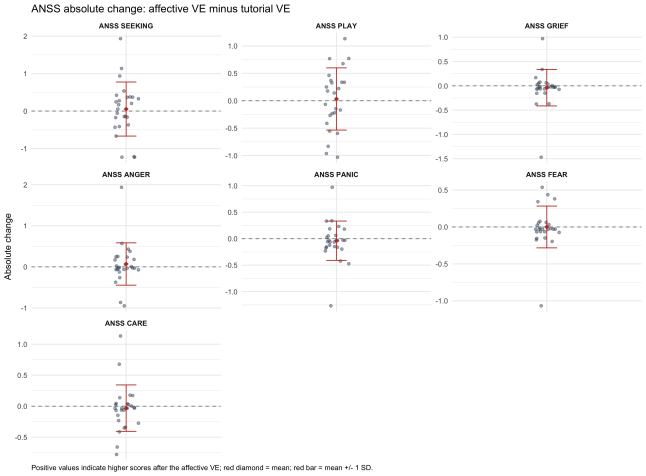
Thus, H2 was not supported: the affective VE did not produce statistically significant increases in SAM, PANAS-X, IPQ, or ANSS scores relative to the post-tutorial VE after Holm correction.

6.2 Post-affective subscale comparisons

To test H3, post-affective Joviality and PLAY were compared with the other subscales within their respective questionnaires. PANAS-X Joviality was higher than Negative Affect

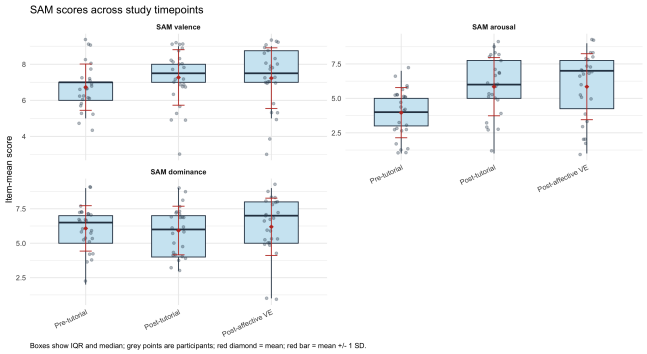


(a) SAM measures

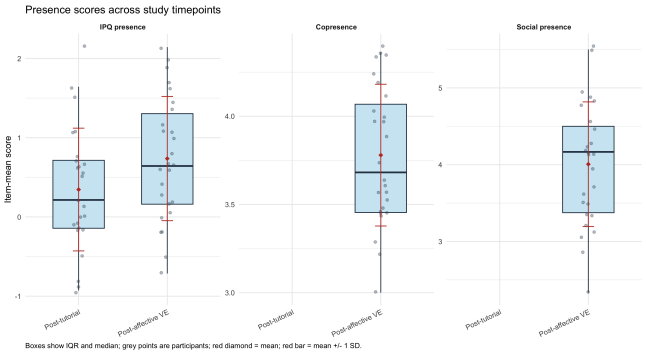


(b) ANSS subscales

Figure 8. Absolute changes between the post-tutorial and post-affective Virtual Environments. Positive values indicate higher scores after the affective VE. Grey points represent participants; red diamonds represent means; red bars represent mean ± one standard deviation.



(a) SAM measures



(b) Presence measures

Figure 9. Questionnaire score distributions across the three study timepoints where available. Boxes show the interquartile range and median; grey points represent individual participants; red diamonds represent means; red bars represent mean ± one standard deviation.

(ΔM = 2.24, W = 297.0, $p_{\text{Holm}} < .001$), Fear (ΔM = 2.18, W = 295.5, $p_{\text{Holm}} < .001$), Hostility (ΔM = 2.32, t = 10.18, $p_{\text{Holm}} < .001$), Guilt (ΔM = 2.30, W = 296.0, $p_{\text{Holm}} < .001$), Sadness (ΔM = 2.28, t = 10.69, $p_{\text{Holm}} < .001$), Self-Assurance (ΔM = 0.73, W = 273.0, $p_{\text{Holm}} < .001$), Shyness (ΔM = 2.10, t = 8.78, $p_{\text{Holm}} < .001$), and Fatigue (ΔM = 1.77, t = 6.12, $p_{\text{Holm}} < .001$). It was not significantly higher than Positive Affect (ΔM = 0.14, t = 1.54, p_{Holm} = .392), Attentiveness (ΔM = 0.14, t = 0.82, p_{Holm} = .421), Serenity (ΔM = 0.40, t = 2.10, p_{Holm} = .187), or Surprise (ΔM = 0.23, t = 1.57, p_{Holm} = .392).

ANSS PLAY was higher than Grief (ΔM = 2.34, t = 11.02, $p_{\text{Holm}} < .001$), Anger (ΔM = 2.30, t = 9.76, $p_{\text{Holm}} < .001$), Panic (ΔM = 2.19, t = 8.85, $p_{\text{Holm}} < .001$), Fear (ΔM = 2.36, t = 10.89, $p_{\text{Holm}} < .001$), and Care (ΔM = 0.42, t = 2.94, p_{Holm} = .014), but not Seeking (ΔM = −0.03, t = −0.29, p_{Holm} = .772). H3 was therefore partially supported: Joviality and PLAY were higher than several, but not all, comparison subscales.

6.3 Post-affective scores relative to predefined thresholds

To test H1, post-affective scores were compared with pre-specified reference points using the post-affective observations only. SAM valence was above the threshold of 6.33 (M = 7.23, SD = 1.68, V = 276.0, p_{Holm} = .042). SAM arousal

($M = 5.85$, $SD = 2.39$, $V = 158.0$, $p_{\text{Holm}} = 1.000$) and dominance ($M = 6.19$, $SD = 2.08$, $V = 191.0$, $p_{\text{Holm}} = 1.000$) were not above the same threshold.

IPQ presence was above its neutral midpoint of zero ($M = 0.74$, $SD = 0.78$, $t = 4.79$, $p_{\text{Holm}} < .001$), and co-presence was above its midpoint of three ($M = 3.78$, $SD = 0.40$, $t = 9.90$, $p_{\text{Holm}} < .001$). Social presence did not differ from its midpoint of four ($M = 4.01$, $SD = 0.81$, $t = 0.04$, $p_{\text{Holm}} = 1.000$). H1 was therefore partially supported: post-affective valence, IPQ presence, and co-presence exceeded their reference points, whereas arousal, dominance, and social presence did not.

6.4 Tutorial baseline check

The pre-tutorial to post-tutorial comparison was a baseline check rather than a test of the affective VE. The tutorial itself changed several measures. SAM arousal increased by 1.88 points ($t = 5.10$, $p_{\text{Holm}} < .001$), PANAS-X Joviality increased by 0.49 points ($W = 316.5$, $p_{\text{Holm}} = .008$), PANAS-X Surprise increased by 1.36 points ($W = 301.5$, $p_{\text{Holm}} = .004$), and ANSS PLAY increased by 0.40 points ($t = 3.89$, $p_{\text{Holm}} = .014$). PANAS-X Fatigue decreased by 0.50 points ($W = 50.5$, $p_{\text{Holm}} = .048$). These findings show that the tutorial was not emotionally neutral for all participants, so the post-tutorial VE remains the appropriate comparator for H2.

6.5 Behavioural indicators

Behavioural indicators were defined as follows: a throw was counted each time the bunny launched an acorn, a successful hit occurred when the bat collided with an acorn and directed it towards the bunny, a catch occurred when the acorn entered the bunny's catch area and an acorn was resolved once it was caught or retrieved after an unsuccessful attempt. Bat swings were counted using the swing-detection event, and bunny-approach attempts were counted when the participant entered the predefined approach area surrounding the bunny. Counts were calculated per participant across the 7.5-minute VE, with catch rate calculated as catches divided by resolved acorns. Successful hits and catches were counted separately because a hit did not always result in a catch.

The bunny generated an average of 79.52 throws ($SD = 6.82$), of which it caught 21.56 acorns on average, corresponding to a catch rate of 0.27 ($SD = 0.16$). Participants made an average of 103.56 acorn-hit attempts ($SD = 36.60$), including 81.04 successful hits ($SD = 26.09$), and made 130.00 bat swings ($SD = 31.43$) and 2.56 bunny-approach attempts ($SD = 2.12$).

Behavioural counts were used descriptively to characterise interaction opportunities. They recorded NPC throws, participant hit attempts, successful hits, catches, bat swings, and bunny approaches as shown in Figure 10. However, timestamps and response latencies were not recorded. Therefore, these counts cannot establish whether participant responses followed NPC actions or whether the NPC's subsequent behaviour was contingent on those responses. Reciprocity was consequently treated as a design intention rather than a directly evaluated outcome.

7 Discussion

This study examined whether the affective Virtual Environment elicited changes in affect, PLAY-related responses, and presence. The results provided limited support for the hypotheses. Post-affective SAM valence scores exceeded the prespecified threshold of 6.33, but arousal and dominance did not. No significant post-tutorial-to-post-affective changes were found for the SAM, PANAS-X, or ANSS measures after Holm correction. In particular, ANSS PLAY changed by only 0.03 points and PANAS-X Joviality by −0.01 points. Direct comparisons also showed that neither target subscale increased more than the other subscales.

The VE may also have engaged the SEEKING system through exploration, anticipation, and goal-directed interaction. SEEKING increased slightly more than PLAY in the primary comparison ($\Delta M = 0.05$ versus 0.03), although this difference was not significant [5, 6]. This suggests that the observed engagement may have reflected general exploratory motivation rather than PLAY specifically. Some participants also reported feeling frustrated when they repeatedly missed the acorns. This may help explain the small descriptive increase in ANSS ANGER, corresponding conceptually to Panksepp's RAGE system, which showed the largest increase among the ANSS subscales ($\Delta M = 0.07$). However, this change was not statistically significant, and the participant comments were not systematically coded, therefore, frustration should be treated as a possible explanation rather than evidence that ANSS ANGER was elicited. The findings may therefore reflect a mixed experience involving PLAY-related social engagement, SEEKING, excitement, and frustration rather than PLAY alone.

Presence-related findings were more positive. IPQ presence and co-presence scores were above their neutral reference points, whereas social presence was not. This suggests that participants generally felt located in the Virtual Environment and perceived the bunny as present, but did not necessarily experience a strong social relationship with it. Behavioural counts showed repeated participant–NPC interaction. However, the absence of timestamps meant that reciprocity and PLAY could not be directly confirmed.

The tutorial itself produced changes in several measures, including arousal, Joviality, Surprise, and PLAY. Therefore, it was not fully emotionally neutral, and increases observed between the pre-tutorial and post-affective assessments cannot be attributed solely to the affective VE. This highlights the importance of interpreting the post-tutorial-to-post-affective comparison as the primary test.

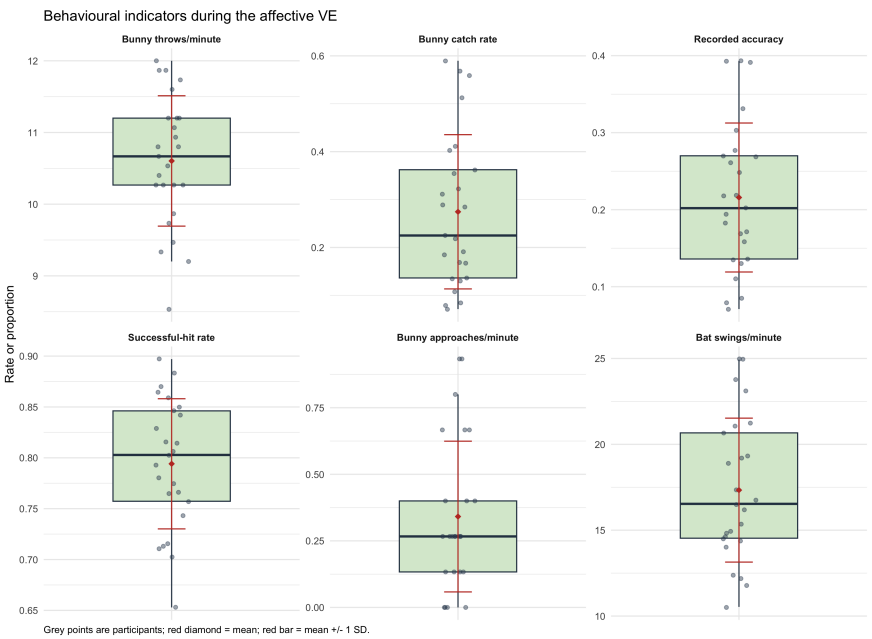


Figure 10. Behavioural indicators during the affective VE. Boxes show the interquartile range and median; red diamonds represent means; red bars represent mean ± one standard deviation.

Several limitations should be considered. The small, snowball-recruited sample limits statistical power and generalisability, as participants may have shared similar social or academic backgrounds. The within-participant design also confounded VE condition with presentation order, familiarity, and novelty. In addition, the limited physical room restricted participants' movement, and contact with or awareness of the boundary may have disrupted presence. Participants also completed the questionnaires after leaving the VE and therefore had to recall their emotional state. Retrospective evaluations can be influenced by particularly intense or final moments and may not accurately represent moment-to-moment affect [10, 16]. Finally, adaptive difficulty may have introduced frustration unrelated to the intended emotional content.

Future studies should use a larger and more diverse sample, counterbalance condition order, provide a larger play area, and collect in-experience or continuous measures. Timestamped behavioural data would also allow the contingent back-and-forth interaction required for reciprocity to be tested directly.

8 Conclusion

This study developed a PLAY-oriented bunny-rally virtual environment and found positive valence, presence, and co-presence, but no selective increase in PLAY-related measures. Because the study used a fixed presentation order and did not record timestamped reciprocal behaviour, the findings demonstrate prototype feasibility rather than confirmed PLAY elicitation

9 Acknowledgments

I would like to thank my supervisor, Professor James Gain, for his guidance and support throughout this project. I also thank my team members, Devin Marais and Zenande Mdludlu, for their contributions to the design framework. I am grateful to James Burness for allowing us to use his tutorial environment. Finally, I acknowledge Garrett LeFever's *Simple Acorn* asset and the bunny-footprint asset sourced from Vecteezy.com.

10 Ethics Statement

Participants' personal information was collected through a Google Form accessible only to project members. Each participant was assigned a randomly generated code, which was used when recording and analysing responses. No personally identifiable information was made publicly available. Another ethical consideration was the use of partial disclosure. Participants were not informed before the study that the experiment specifically aimed to elicit PLAY, as this knowledge could have influenced their behaviour and self-reported emotional responses. This withholding of the specific hypothesis was considered ethically acceptable because it was necessary to preserve the study's validity, posed no more than minimal risk, was approved by the UCT Inter-faculty Research Ethics Committee (IFHREC/02979/2026), and participants were fully debriefed immediately after the study.

Participants were informed of the study's focus during debriefing.

11 AI declaration

ChatGPT was used to generate the floor-plan image and to assist with editing and proofreading of the manuscript. The authors reviewed and verified all AI-assisted content.

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A Supplementary information

The tables report the test selected for each measure after inspection of the Shapiro–Wilk test and Q–Q plot. p is the unadjusted two-sided value and p_{Holm} is the Holm-adjusted value within the corresponding comparison family. W denotes the Wilcoxon signed-rank statistic for paired comparisons; V denotes the one-sample Wilcoxon statistic for threshold comparisons.

Table 3. Primary post-affective minus post-tutorial comparisons.

Measure	n	Test	Statistic	p	p_{Holm}
SAM valence	26	Wilcoxon	$W = 70.5$	.915	1.000
SAM arousal	26	Wilcoxon	$W = 72.0$	.854	1.000
SAM dominance	26	paired t	$t = .78$	.442	1.000
PANAS-X Negative Affect	24	Wilcoxon	$W = 61.0$	.295	1.000
PANAS-X Positive Affect	24	paired t	$t = .00$	1.000	1.000
PANAS-X Fear	24	Wilcoxon	$W = 36.0$	.068	1.000
PANAS-X Hostility	24	Wilcoxon	$W = 32.5$	.633	1.000
PANAS-X Guilt	24	Wilcoxon	$W = 32.5$	1.000	1.000
PANAS-X Sadness	24	Wilcoxon	$W = 13.0$	.044	1.000
PANAS-X Joviality	24	paired t	$t = -.04$	.969	1.000
PANAS-X Self-Assurance	24	paired t	$t = -.28$	.779	1.000
PANAS-X Attentiveness	24	Wilcoxon	$W = 157.5$	.318	1.000
PANAS-X Shyness	24	paired t	$t = -1.82$	.081	1.000
PANAS-X Fatigue	24	paired t	$t = -.31$	.758	1.000
PANAS-X Serenity	24	paired t	$t = .50$	.625	1.000
PANAS-X Surprise	24	Wilcoxon	$W = 88.5$	.463	1.000
ANSS SEEKING	26	paired t	$t = .38$	.707	1.000
ANSS PLAY	25	paired t	$t = .28$	.780	1.000
ANSS GRIEF	26	Wilcoxon	$W = 13.0$	.526	1.000
ANSS ANGER	26	Wilcoxon	$W = 64.0$	.484	1.000
ANSS PANIC	26	Wilcoxon	$W = 55.5$	.531	1.000
ANSS FEAR	26	Wilcoxon	$W = 26.0$	.720	1.000
ANSS CARE	26	Wilcoxon	$W = 34.5$	.458	1.000
IPQ presence	25	paired t	$t = 2.39$	.025	.597

Table 4. Pre-tutorial minus post-tutorial baseline comparisons.

Measure	n	Test	Statistic	p	p_{Holm}
SAM valence	26	Wilcoxon	$W = 149.0$	.099	1.000
SAM arousal	26	paired t	$t = 5.10$	$< .001$	$< .001$
SAM dominance	26	Wilcoxon	$W = 63.0$	.809	1.000
PANAS-X Negative Affect	26	Wilcoxon	$W = 83.5$	.273	1.000
PANAS-X Positive Affect	26	Wilcoxon	$W = 271.5$	.003	.062
PANAS-X Fear	26	Wilcoxon	$W = 76.0$	.662	1.000
PANAS-X Hostility	26	Wilcoxon	$W = 44.0$	.376	1.000
PANAS-X Guilt	26	Wilcoxon	$W = 22.0$	.610	1.000
PANAS-X Sadness	26	Wilcoxon	$W = 37.0$	.113	1.000
PANAS-X Joviality	26	Wilcoxon	$W = 316.5$	$< .001$	.008
PANAS-X Self-Assurance	26	paired t	$t = .04$	.967	1.000
PANAS-X Attentiveness	26	paired t	$t = .78$	.446	1.000
PANAS-X Shyness	26	Wilcoxon	$W = 79.0$	.338	1.000
PANAS-X Fatigue	26	Wilcoxon	$W = 50.5$	.003	.048
PANAS-X Serenity	26	paired t	$t = -3.20$	.004	.070
PANAS-X Surprise	26	Wilcoxon	$W = 301.5$	$< .001$	.004
ANSS SEEKING	26	Wilcoxon	$W = 236.5$	.048	.762
ANSS PLAY	26	paired t	$t = 3.89$	$< .001$	.014
ANSS GRIEF	26	Wilcoxon	$W = 30.5$	.309	1.000
ANSS ANGER	26	Wilcoxon	$W = 24.5$	.265	1.000
ANSS PANIC	26	Wilcoxon	$W = 37.0$	.064	.967
ANSS FEAR	26	Wilcoxon	$W = 27.0$	.115	1.000
ANSS CARE	25	paired t	$t = -.65$	.525	1.000

Table 5. Post-affective subscale comparisons for H3.

Comparison	n	Test	Statistic	p_{Holm}
Joviality vs Negative Affect	24	Wilcoxon	$W = 297.0$	$< .001$
Joviality vs Positive Affect	24	paired t	$t = 1.54$	.392
Joviality vs Fear	24	Wilcoxon	$W = 295.5$	$< .001$
Joviality vs Hostility	24	paired t	$t = 10.18$	$< .001$
Joviality vs Guilt	24	Wilcoxon	$W = 296.0$	$< .001$
Joviality vs Sadness	24	paired t	$t = 10.69$	$< .001$
Joviality vs Self-Assurance	24	Wilcoxon	$W = 273.0$	$< .001$
Joviality vs Attentiveness	24	paired t	$t = .82$	.421
Joviality vs Shyness	24	paired t	$t = 8.78$	$< .001$
Joviality vs Fatigue	24	paired t	$t = 6.12$	$< .001$
Joviality vs Serenity	24	paired t	$t = 2.10$	.187
Joviality vs Surprise	24	paired t	$t = 1.57$	.392
PLAY vs SEEKING	25	paired t	$t = -.29$	.772
PLAY vs GRIEF	25	paired t	$t = 11.02$	$< .001$
PLAY vs ANGER	25	paired t	$t = 9.76$	$< .001$
PLAY vs PANIC	25	paired t	$t = 8.85$	$< .001$
PLAY vs FEAR	25	paired t	$t = 10.89$	$< .001$
PLAY vs CARE	25	paired t	$t = 2.94$	.014

Table 6. Post-affective scores relative to predefined thresholds for H1.

Measure	n	Threshold	Test	Statistic	p	p_{Holm}
SAM valence	26	6.33	Wilcoxon	$V = 276.0$	.011	.042
SAM arousal	26	6.33	Wilcoxon	$V = 158.0$	.664	1.000
SAM dominance	26	6.33	Wilcoxon	$V = 191.0$	.702	1.000
IPQ presence	26	0	paired t	$t = 4.79$	$< .001$	$< .001$
Co-presence	26	3	paired t	$t = 9.90$	$< .001$	$< .001$
Social presence	26	4	paired t	$t = .04$	.968	1.000